

A Generalized DPO-Based Frame Selection Framework for Efficient Video Understanding

1. Introduction

Over the past decade, deep learning architectures, including CNNs, Vision Transformers (ViTs), VideoMAE, YOLO, and CLIP have transformed computer vision by learning directly from large-scale datasets. Driven by powerful computational hardware, these models have become the foundation for critical applications such as object detection, human action recognition, autonomous driving, and video understanding.

Despite substantial structural variations, these pipelines generally operate under a uniform, resource-intensive assumption: every incoming video frame carries crucial semantic information and must be processed. While this simplifies architecture design, it rarely reflects the redundant nature of real-world video streams.

1.1 Problem Statement

Current video architectures process every frame uniformly, causing massive semantic redundancy that inflates GPU compute, latency, and power consumption, rendering edge deployment impractical. Traditional remedies like heuristic sampling or reinforcement learning fail due to strict architecture dependence, volatile optimization, and complex reward engineering. To resolve this, a model-independent, generalized preprocessing framework is required to intelligently filter out redundant frames before downstream inference occurs.

2. Literature Review Summary

Prior attempts to mitigate frame redundancy typically balance on two paradigms: handcrafted heuristics or reinforcement learning (RL) agents. Heuristic sampling (e.g., uniform stride or color histogram variance) provides low-overhead execution but suffers from an acute lack of task-specific semantic understanding. Conversely, RL-driven adaptive models frame selection as a sequential Markov Decision Process. While semantically aware, RL methods are heavily restricted by high hyperparameter sensitivity, unstable training gradients, architecture dependency, and the challenging requirement of manually tuning complex numerical reward functions.

2.1 Comparison of Existing Frame Selection Approaches

Method Category	Representative Methods	Learning Strategy	Semantic Awareness	Reward Function	Generalization	Major Limitations
Fixed Frame Sampling	Uniform Sampling	Rule-Based	Low	Not Required	High	Important events may be skipped
Key-Frame Extraction	Optical Flow, Scene Detection	Handcrafted	Low	Not Required	Moderate	Handcrafted heuristics
Reinforcement Learning	PPO, DQN, A2C	Sequential Decision Making	High	Required	Moderate	Reward engineering and unstable optimization
Proposed Framework	DPO-Based Frame Selection	Preference Learning	High	Not Required	High	Requires preference pair generation and experimental validation

3. Proposed DPO-Based Frame Selection Framework

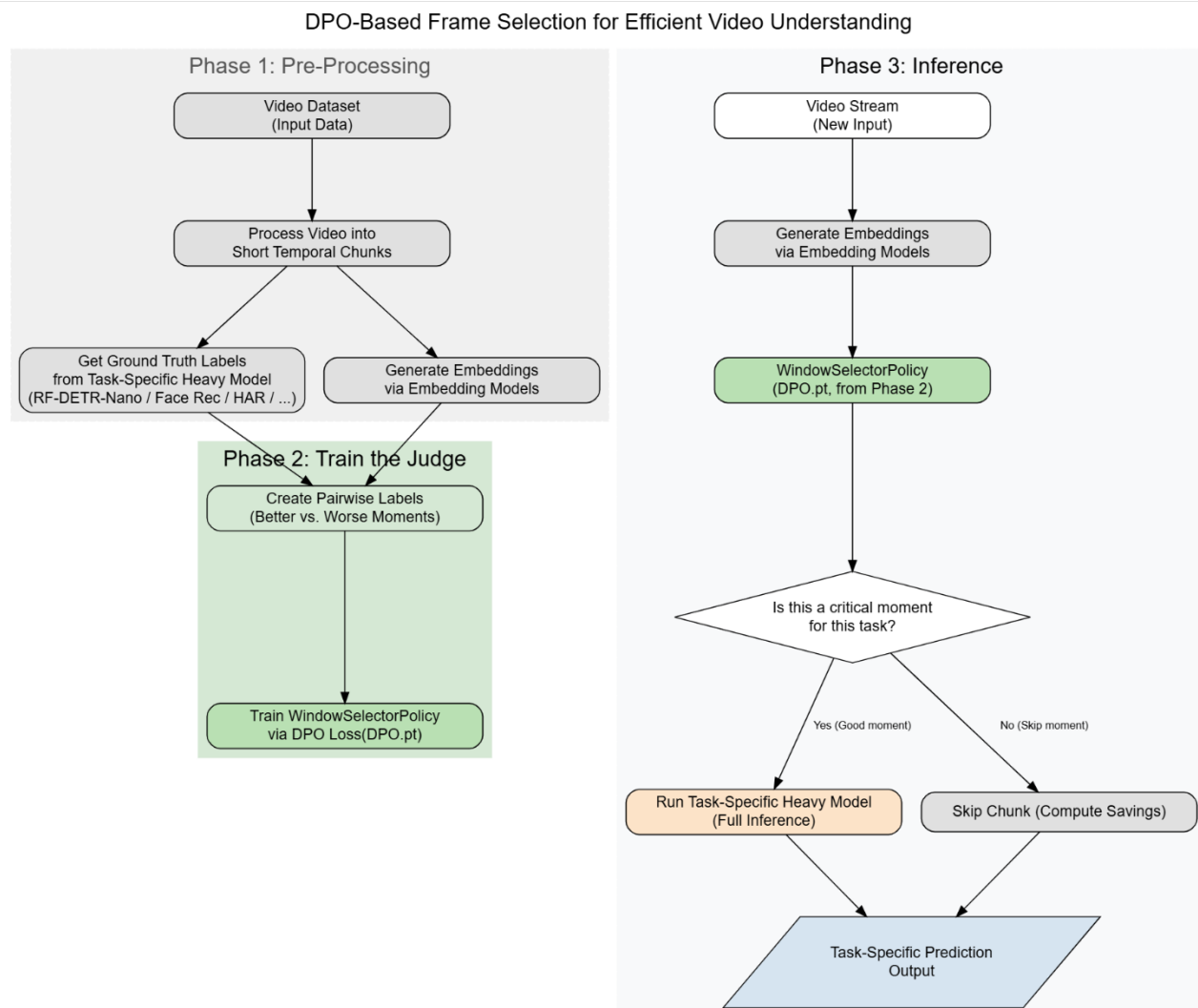
3.1 Introduction

This research introduces a model-independent, Direct Preference Optimization (DPO)-based frame selection framework that reformulates filtering as a preference learning problem. Rather than using volatile reinforcement learning rewards, a capable teacher model ranks frame pairs to capture semantic relationships. These preferences train a lightweight DPO policy to determine whether incoming frames are processed or skipped before expensive downstream inference, allowing the identical policy architecture to adapt across diverse vision tasks by changing only the embedding extractor and teacher model.

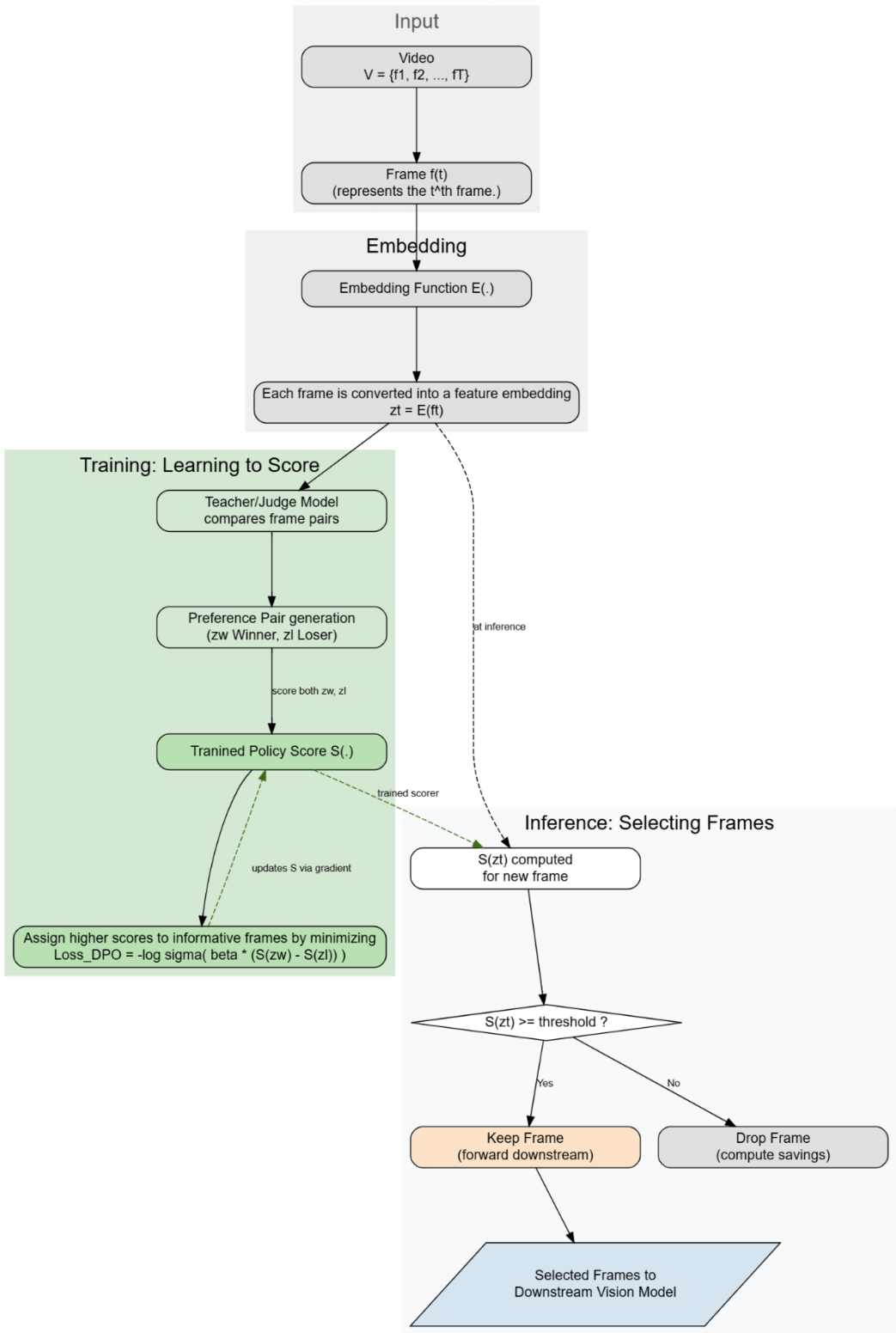
3.2 Framework Overview

The framework operates across three major sequential phases:

- **Data Preparation**
- **DPO Policy Training**
- **Inference**



Working of DPO(Direct Preference Optimization)



Downstream Application	Teacher Model Archetype
Human Action Recognition	VideoMAE
Object Detection	RF-DETR / YOLO
Face Recognition	ArcFace
Pose Estimation	ViTPose
Person Re-ID	TransRelD
Video Captioning	BLIP-2

Target Application	Embedding Extractor	Teacher Model	Downstream Vision Model
Human Action Recognition	VideoMAE Encoder	VideoMAE	VideoMAE, SlowFast
Object Detection	DINOv2	RF-DETR, YOLO	RF-DETR, YOLO
Face Recognition	ArcFace	ArcFace	ArcFace, FaceNet
Face Verification	ArcFace	ArcFace	ArcFace
Person Re-identification	DINOv2 / CLIP	TransRelD	TransRelD
Pose Estimation	DINOv2	ViTPose	ViTPose
Gesture Recognition	VideoMAE	VideoMAE	VideoMAE
Video Captioning	CLIP	BLIP-2	BLIP-2
Video Question Answering	CLIP	LLaVA-Video / Qwen2.5-VL	LLaVA-Video / Qwen2.5-VL

3.2.1 Framework Scalability

Since the frame selection policy operates fully independent of the downstream model, future computer vision architectures can be integrated by simply replacing the embedding extractor and teacher model. If a new state-of-the-art model emerges, it can serve as the teacher while the same optimization pipeline is retained.

3.3 Advantages of the Proposed Framework

- Reward-Free Optimization
- Stable Training
- Model-Independent Design
- Reduced Computational Cost
- Improved Real-Time Performance
- Edge AI Compatibility

3.4 Comparison with Reinforcement Learning

Feature	Proximal Policy Optimization (PPO)	Proposed DPO Framework
Learning Paradigm	Reinforcement Learning	Preference Learning
Reward Function	Required	Not Required
Manual Reward Design	Required	Not Required
Optimization Strategy	Reward Maximization	Pairwise Preference Optimization
Training Stability	Moderate	High (Expected)
Hyperparameter Sensitivity	High	Lower
Sample Efficiency	Moderate	Higher (Expected)
Training Complexity	High	Moderate
Policy Learning	Indirect	Direct
Generalized Framework	Limited	Yes

4. Future Work

Several empirical vectors remain for future investigative scaling:

- **Dynamic Threshold Selection:** Investigating adaptive mechanisms for shifting the selection threshold dynamically during runtime inference.
- **Multi-Scale Temporal Windows:** Expanding selection boundaries beyond discrete static frames to variable multi-scale temporal video windows.
- **Self-Supervised Preference Generation:** Mitigating teacher model dependency by generating automated, self-supervised preference pairs.
- **Online & Multi-Modal Selection:** Allowing the policy to adapt continuously to streaming shifts, and integrating complementary language or audio signals alongside raw visual tokens.
- **Foundation Integration & Edge Deployment:** Scaling the framework to act as a frontend filter for large-scale video foundation models and evaluating execution speeds on small embedded edge AI compute hardware.

5. Conclusion

This report presented a generalized Direct Preference Optimization (DPO)-based frame selection framework targeted at driving efficient video understanding. By steering clear of manual reward modeling, the framework trains lightweight selection policies using robust, pairwise teacher supervision. Designed entirely as a model-independent preprocessing component, it offers modular compatibility across a wide matrix of vision tasks by simply interchanging embedding extractors and teacher pairs. While definitive empirical validation remains the critical next step, this framework establishes a comprehensive, mathematically grounded foundation for exploring preference learning as a viable, scalable alternative to reinforcement learning in video processing systems.

